

Physics 828: Homework Set No. 5

**Due date: Friday, February 17, 2011, 1:00pm
in PRB M2043 (Biao Huang's office)**

Total point value of set: 60 points

Problem 1 (10 pts.): Exercise 12.5.12 (Shankar, p. 338)

Problem 2 (10 pts.): Exercise 12.5.13 (Shankar, p. 338)

Problem 3 (10 pts.): Exercise 12.6.1 (Shankar, p. 340)

Problem 4 (10 pts.): Exercise 12.6.8 (Shankar, p. 349)

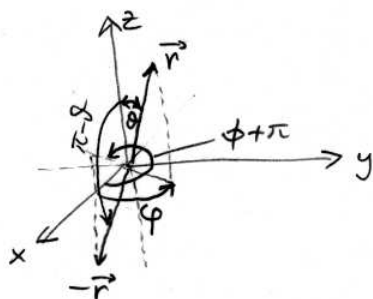
Problem 5 (20 pts.): Exercise 12.6.9 (Shankar, p. 349)

Exercise 12.5.12 (p. 338)

$[\hat{L}^2, \hat{\pi}] = [\hat{L}_z, \hat{\pi}] = 0 \Rightarrow$ common set of eigenstates for $\hat{L}^2, \hat{L}_z$ and $\hat{\pi}$.

Since $|lm\rangle \leftrightarrow \langle \vartheta, \varphi | lm \rangle = Y_{lm}(\vartheta, \varphi)$ completely classifies the common eigenstates of $\hat{L}^2, \hat{L}_z$, with no remaining degeneracy, the states $|lm\rangle$ should automatically be eigenstates of $\hat{\pi}$.

Under parity $\hat{\pi} : \vec{r} \mapsto -\vec{r}$
 $(x, y, z) \mapsto (-x, -y, -z)$



$$\begin{aligned} r \sin \vartheta \cos \varphi &\mapsto -r \sin \vartheta \cos \varphi = r \sin(\pi - \vartheta) \cos(\varphi + \pi) \\ r \sin \vartheta \sin \varphi &\mapsto -r \sin \vartheta \sin \varphi = r \sin(\pi - \vartheta) \sin(\varphi + \pi) \\ r \cos \vartheta &\mapsto -r \cos \vartheta = r \cos(\pi - \vartheta) \end{aligned}$$

hence $\vartheta \xrightarrow{\hat{\pi}} \pi - \vartheta$
 $\varphi \mapsto \varphi + \pi \quad \checkmark$

$$\begin{aligned} Y_{\ell\ell}(\vartheta, \varphi) &= (-1)^\ell \sqrt{\frac{(2\ell+1)!}{4\pi}} \frac{1}{2^\ell \ell!} \sin^\ell(\vartheta) e^{i\ell\varphi} \\ &\xrightarrow{\hat{\pi}} (-1)^\ell \sqrt{\frac{(2\ell+1)!}{4\pi}} \frac{1}{2^\ell \ell!} \frac{\sin^\ell(\pi - \vartheta)}{= \sin^\ell \vartheta} \underbrace{e^{i\ell(\varphi + \pi)}}_{= (e^{i\pi})^\ell e^{i\ell\varphi}} = (-1)^\ell e^{i\ell\varphi} \\ &= (-1)^\ell Y_{\ell\ell}(\vartheta, \varphi) \end{aligned}$$

Next $Y_{\ell, m-1}(\vartheta, \varphi) \sim L_- Y_{\ell m}(\vartheta, \varphi) = -\hbar e^{-i\varphi} \left(\frac{\partial}{\partial \vartheta} - i \frac{\cos \vartheta}{\sin \vartheta} \frac{\partial}{\partial \varphi} \right) Y_{\ell m}$

Under the transformation $\vartheta \rightarrow \pi - \vartheta, \varphi \rightarrow \varphi + \pi$

$$\begin{aligned} L_- &\mapsto -\hbar e^{-i\varphi} \underbrace{e^{-i\pi}}_{(-1)} \left(\underbrace{\frac{\partial}{\partial(\pi - \vartheta)}}_{= -\frac{\partial}{\partial \vartheta}} - i \frac{\cos(\pi - \vartheta)}{\sin(\pi - \vartheta)} \underbrace{\frac{\partial}{\partial(\varphi + \pi)}}_{= \frac{\partial}{\partial \varphi}} \right) \\ &= -\hbar e^{-i\varphi} \left(\frac{\partial}{\partial \vartheta} - i \cot \vartheta \frac{\partial}{\partial \varphi} \right) = L_- \Rightarrow \end{aligned}$$

$$\Rightarrow [\hat{L}_-, \hat{\pi}] = 0$$

So we can write the following chain of arguments:

$$\underline{l=m}: \hat{\pi} |ll\rangle = (-1)^l |ll\rangle$$

$$\underline{m \rightarrow m-1}: \hat{\pi} |l, m-1\rangle = \hat{\pi} \frac{1}{\hbar \sqrt{(l+m)(l-m+1)}} \hat{L}_- |lm\rangle$$

$$= \frac{\hat{L}_-}{\hbar \sqrt{(l+m)(l-m+1)}} \underbrace{\hat{\pi} |lm\rangle}_{= (-1)^l |lm\rangle}$$

$$= (-1)^l \frac{\hat{L}_-}{\hbar \sqrt{(l+m)(l-m+1)}} |lm\rangle = \underline{(-1)^l |l, m-1\rangle}$$

Hence $\boxed{\hat{\pi} |lm\rangle = (-1)^l |lm\rangle \quad \forall l, m}$

Exercise 12.5.13 (p. 338)

$$\begin{aligned}\psi(x, y, z) &= N(x + y + iz) e^{-\alpha r} & r &= \sqrt{x^2 + y^2 + z^2} \\ &= N(r \sin \vartheta (\sin \varphi + i \cos \varphi) + 2r \cos \vartheta) e^{-\alpha r} \\ &= N r e^{-\alpha r} (\sin \vartheta (\sin \varphi + i \cos \varphi) + 2 \cos \vartheta)\end{aligned}$$

(1)

$$\begin{aligned}Y_{1,\pm 1}(\vartheta, \varphi) &= \mp \sqrt{\frac{3}{8\pi}} \sin \vartheta e^{\pm i\varphi} & (12.5.39) \\ &= \mp \sqrt{\frac{3}{8\pi}} \frac{r \sin \vartheta (\cos \varphi \pm i \sin \varphi)}{r} \\ &= \mp \sqrt{\frac{3}{4\pi}} \frac{x \pm iy}{\sqrt{2} r}\end{aligned}$$

$$Y_{10}(\vartheta, \varphi) = \sqrt{\frac{3}{4\pi}} \cos \vartheta = \sqrt{\frac{3}{4\pi}} \frac{z}{r}$$

$$\Rightarrow Y_{11} + Y_{1,-1} = \sqrt{\frac{3}{8\pi}} \left(\frac{-x - iy}{r} + \frac{x - iy}{r} \right) = -2i \frac{y}{r} \sqrt{\frac{3}{8\pi}}$$

$$\Rightarrow y = i \sqrt{\frac{8\pi}{3}} r \frac{Y_{11} + Y_{1,-1}}{2} = i \sqrt{\frac{4\pi}{3}} r \left(\frac{Y_{1,-1} + Y_{11}}{\sqrt{2}} \right)$$

$$Y_{11} - Y_{1,-1} = \sqrt{\frac{3}{8\pi}} \left(\frac{-x - iy}{r} - \frac{x - iy}{r} \right) = -2 \sqrt{\frac{3}{8\pi}} \frac{x}{r}$$

$$\Rightarrow x = -\sqrt{\frac{8\pi}{3}} r \frac{Y_{1,-1} - Y_{11}}{2} = \sqrt{\frac{4\pi}{3}} r \left(\frac{Y_{1,-1} - Y_{11}}{\sqrt{2}} \right)$$

$$z = \sqrt{\frac{4\pi}{3}} r Y_{10}$$

$$(2) \quad \Rightarrow x + y + iz = \sqrt{\frac{8\pi}{3}} r \left(\frac{Y_{1,-1} - Y_{11}}{2} + i \frac{Y_{1,-1} + Y_{11}}{2} + \frac{2}{\sqrt{2}} Y_{10} \right)$$

$$= \sqrt{\frac{4\pi}{3}} r \left(Y_{1,-1} \frac{1+i}{\sqrt{2}} + Y_{11} \frac{i-1}{\sqrt{2}} + 2 Y_{10} \right)$$

$$= (c(\ell_2=-1) Y_{1,-1} + c(\ell_2=1) Y_{11} + d(\ell_2=0) Y_{10}) \sqrt{\frac{4\pi}{3}} r$$

The probabilities for finding a given value of l_z are $\sim |c(l_z)|^2$

$$|c(l_z=0)|^2 = 4$$

$$|c(l_z=1)|^2 = \frac{i-1}{\sqrt{2}} \cdot \frac{-i-1}{\sqrt{2}} = \frac{1+i-i+1}{2} = 1$$

$$|c(l_z=-1)|^2 = \frac{(1+i)(1-i)}{2} = 1$$

These add up to 6; hence

$$P(l_z=0) = \frac{4}{6} = \frac{2}{3}$$

$$P(l_z=-1) = \frac{1}{6} = P(l_z=+1)$$

Exercise 12.6.1 (p.340)

$$\psi_E(r, \vartheta, \varphi) = A e^{-r/a_0} \Rightarrow u_E(r) = r e^{-r/a_0}$$

$$(1) \psi_E \text{ is independent of } \vartheta, \varphi \Rightarrow \psi_E(r, \vartheta, \varphi) \sim e^{-r/a_0} Y_{00}(\vartheta, \varphi)$$

$\Rightarrow$ The state $|\psi_E\rangle$ is an angular momentum eigenstate with $l=0$ and $m=0$.

$$(2) \text{ Assume } V(r) \xrightarrow{r \rightarrow \infty} 0$$

Schrödinger equation:

$$\left(\frac{d^2}{dr^2} + \frac{2\mu}{\hbar^2} (E - V(r)) \right) r e^{-r/a_0} = 0$$

$$\begin{aligned} \frac{d^2}{dr^2} (r e^{-r/a_0}) &= \frac{d}{dr} \left(-\frac{r}{a_0} e^{-r/a_0} + e^{-r/a_0} \right) \\ &= -\frac{1}{a_0} e^{-r/a_0} + \frac{r}{a_0^2} e^{-r/a_0} - \frac{1}{a_0} e^{-r/a_0} \\ &= \frac{1}{a_0} \left(\frac{r}{a_0} - 2 \right) e^{-r/a_0} \end{aligned}$$

$$\Rightarrow \frac{1}{a_0} \left(\frac{r}{a_0} - 2 \right) e^{-r/a_0} = -\frac{2\mu E}{\hbar^2} r e^{-r/a_0} + \frac{2\mu}{\hbar^2} V(r) r e^{-r/a_0}$$

$$\Rightarrow \frac{1}{a_0} \left(\frac{1}{a_0} - \frac{2}{r} \right) = -\frac{2\mu E}{\hbar^2} + \frac{2\mu}{\hbar^2} V(r)$$

Take $r \rightarrow \infty$, use $V(r) \rightarrow 0$:

$$\frac{1}{a_0^2} = -\frac{2\mu E}{\hbar^2} \Rightarrow \boxed{E = -\frac{\hbar^2}{2\mu a_0^2}}$$

(3) Back to the full Schrödinger equation:

(remember, $l=0$, hence no centrifugal potential)

$$\frac{1}{a_0^2} - \frac{2}{ra_0} = + \frac{1}{a_0^2} + \frac{2\mu}{\hbar^2} V(r)$$

$$\Rightarrow \boxed{V(r) = \frac{\hbar^2}{2\mu} \left(-\frac{2}{a_0 r} \right) = -\frac{\hbar^2}{\mu a_0 r}}$$

Exercise 12.6.8 (p. 349)

Spherical box of radius r_0 : $V(r) = \lim_{V_0 \rightarrow \infty} V_0 \Theta(r-r_0)$

Schrödinger equation:

$$\Psi_{Elm}(r, \vartheta, \varphi) = \frac{u_{El}(r)}{r} Y_{lm}(\vartheta, \varphi)$$

$$\left[\frac{d^2}{dr^2} + \underbrace{\frac{2\mu}{\hbar^2} (E - V(r))}_{k^2} - \frac{l(l+1)}{r^2} \right] u_{El}(r) = 0$$

$$\text{For } r < r_0: \left[\frac{d^2}{dr^2} + k^2 - \frac{l(l+1)}{r^2} \right] u_{El}(r) = 0$$

$$\text{For } r \geq r_0: u_{El}(r) \equiv 0$$

Change variables $\rho = kr$

$$\left(-\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2} \right) u_{El}(\rho) = 0$$

$\Rightarrow$ solution is that for a free particle, but with a boundary condition: $u_{El}(r_0) = 0$

The solution must be regular at $r=0$

$$\Rightarrow \Psi_{nlm}(r, \vartheta, \varphi) = N j_l(k_n r) Y_{lm}(\vartheta, \varphi)$$

with $k_n r_0 = n^{\text{th}} \text{ zero of } j_l(x)$

$$\text{For } l=0: j_0(\rho) = \frac{\sin \rho}{\rho}$$

$$\Rightarrow \psi_{n00}(r) = N_n \frac{\sin(k_n r)}{r} \quad \text{with } k_n r_0 = n\pi \quad (n=1, 2, 3, \dots)$$

$$\Rightarrow k_n^2 r_0^2 = n^2 \pi^2 \Rightarrow \frac{2\mu r_0^2}{\hbar^2} E_n = n^2 \pi^2$$

$$\Rightarrow \boxed{E_n = \frac{\hbar^2 n^2 \pi^2}{2\mu r_0^2}}$$

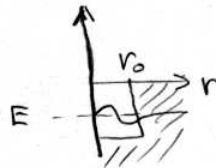
$\mu = \text{mass of particle}$

Exercise 12.6.9 (p. 349)

$$V(r) = -V_0 \Theta(r-r_0) \quad \text{spherical well}$$

$$\Psi_E(r, \vartheta, \varphi) = \frac{U_E(r)}{r} \quad \text{for } l=0 \text{ states}$$

$$-V_0 < E < 0 \quad \text{for bound states:}$$



Schrödinger equation for $l=0$:

$$\left(\frac{d^2}{dr^2} + \frac{2\mu(V_0 - |E|)}{\hbar^2} \right) U_E(r) = 0 \quad r < r_0$$

$$\left(\frac{d^2}{dr^2} - \frac{2\mu|E|}{\hbar^2} \right) U_E(r) = 0 \quad r \geq r_0$$

$$\text{define } \kappa^2 = \frac{2\mu|E|}{\hbar^2} > 0, \quad k^2 = \frac{2\mu(V_0 - |E|)}{\hbar^2} > 0$$

$$\Rightarrow U_E(r) = \begin{cases} A \sin(kr) + B \cos(kr) & r < r_0 \\ C e^{-\kappa r} + D e^{\kappa r} & r \geq r_0 \end{cases}$$

$$\text{Normalizability} \rightarrow B=0, D=0$$

At $r=r_0$: Continuity of ψ :

$$A \sin(kr_0) = C e^{-\kappa r_0}$$

Continuity of ψ' :

$$A k \cos(kr_0) = -\kappa C e^{-\kappa r_0}$$

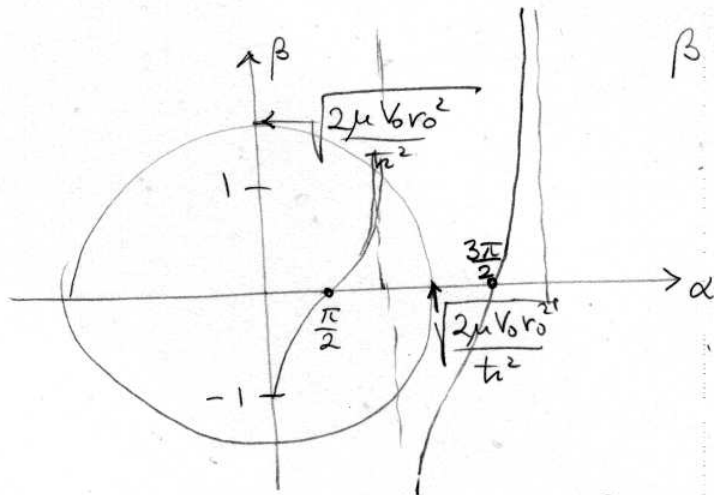
Divide by each other:

$$\frac{1}{k} \tan(kr_0) = -\frac{1}{\kappa} \Rightarrow \boxed{\frac{k}{\kappa} = -\tan(kr_0)} \quad \checkmark$$

$$\text{or } \frac{\kappa r_0}{k r_0} = -\cot(kr_0)$$

Define $\beta = \kappa r_0$, $\alpha = k r_0$

$$\alpha^2 + \beta^2 = \frac{2\mu r_0^2 V_0}{\hbar^2} \quad \text{and} \quad \beta = -\alpha \cot \alpha$$



$$\beta = 0 \text{ at } \cos \alpha = 0 \Leftrightarrow \alpha = (2n+1)\frac{\pi}{2}$$

Only $\alpha > 0, \beta > 0$ are physical

If $\sqrt{\frac{2\mu r_0^2 V_0}{\hbar^2}} < \frac{\pi}{2}$, then the two curves do not intersect in the first quadrant $\rightarrow$ no bound state.

$\Rightarrow$ No bound state if

$$\boxed{V_0 < \frac{\pi^2 \hbar^2}{8\mu r_0^2}} \checkmark$$

If V_0 is large, the circle above is big and intersects with several branches of the $\alpha \cot \alpha$ function $\rightarrow$ several bound states.